



INSTITUTE OF ENGINEERING & TECHNOLOGY

MOHANLAL SUKHADIA UNIVERSITY, UDAIPUR

Name : Lavina Tamboli | Class: Btech-CSE IV sem) | Subject: DBMS Lab

Experiment-2

Objective: Apply the constraints like Primary key, Foreign key, NOT NULL to the table.

Step-1: Write a Query to show Primary key, foreign key.

```
mysql> create table employee_lavina(employee_id int primary key, first_name varchar(20), last_name varchar(20), salary int, department_id int, hire date, CONSTRAINT fk_dept FOREIGN KEY (department_id) REFERENCES departments(department_id));
Query OK, 0 rows affected (0.04 sec)
```

Step-2: Write a Query to show the table name Employee_lavina.

```
mysql> select * from employee_lavina;
+-----+-----+-----+-----+-----+-----+
| employee_id | first_name | last_name | salary | department_id | hire       |
+-----+-----+-----+-----+-----+-----+
| 1           | John      | Smith    | 60000  | 1             | 2022-01-15 |
| 2           | Jane      | Doe       | 15000  | 1             | 2021-03-20 |
| 3           | Peter     | Jones    | 90000  | 2             | 2020-05-10 |
| 4           | Mary     | Williams | 80000  | 2             | 2021-08-01 |
| 5           | David     | Brown    | 65000  | 3             | 2023-10-05 |
| 6           | null      | Davis    | 55000  | 3             | 2023-02-12 |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Step-3: Write a Query to show table name Departments.

```
mysql> select * from departments;
+-----+-----+
| department_id | department_name |
+-----+-----+
| 1             | Engineering     |
| 2             | Sales           |
| 3             | Marketing       |
| 4             | HR              |
+-----+-----+
4 rows in set (0.00 sec)
```

Step-4: Write a Query to display employee first name and salary sorted from highest salary to lowest salary.



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```
mysql> select * from employee_lavina order by salary desc;
```

employee_id	first_name	last_name	salary	department_id	hire
3	Peter	Jones	90000	2	2020-05-10
4	Mary	Williams	80000	2	2021-08-01
5	David	Brown	65000	3	2023-10-05
1	John	Smith	60000	1	2022-01-15
6	null	Davis	55000	3	2023-02-12
2	Jane	Doe	15000	1	2021-03-20

6 rows in set (0.00 sec)

Step-5: Write a Query to display the three employee having highest salary from table employee_lavina.

```
mysql> select * from employee_lavina order by salary desc limit 3;
```

employee_id	first_name	last_name	salary	department_id	hire
3	Peter	Jones	90000	2	2020-05-10
4	Mary	Williams	80000	2	2021-08-01
5	David	Brown	65000	3	2023-10-05

3 rows in set (0.00 sec)

Step-6: Write a Query to display the employee id from employee_lavina where first name is NULL.

```
mysql> select employee_id from employee_lavina where first_name is null;
```

employee_id
6

1 row in set (0.00 sec)

Step-7: Write a Query to display different department id from table employee_lavina.

```
mysql> select distinct department_id from employee_lavina;
```

department_id
1
2
3

3 rows in set (0.01 sec)

Step-8: Write a Query to Written the total number of employee in the organisation.

```
mysql> select count(employee_id) from employee_lavina;
```

count(employee_id)
6

1 row in set (0.01 sec)



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Step-9: Write a Query to sum salary of all employee from the table employee_lavina.

```
mysql> select sum(salary) from employee_lavina;
+-----+
| sum(salary) |
+-----+
|      365000 |
+-----+
1 row in set (0.01 sec)
```

Step-10: Write a Query to calculate the total salary of all the employee in the Engineering department.

```
mysql> select sum(salary) as total_salary
-> from employee_lavina as E, departments as D
-> where E.department_id = D.department_id and department_name = "Engineering";
+-----+
| total_salary |
+-----+
|      75000 |
+-----+
1 row in set (0.01 sec)
```

Step-11: Write a Query to calculate the average of the total salary of all the employee .

```
mysql> select avg(salary) from employee_lavina;
+-----+
| avg(salary) |
+-----+
|  60833.3333 |
+-----+
1 row in set (0.00 sec)
```

Step-12: Write a Query to show how many employee are there in the each departments.

```
mysql> select department_name, count(employee_id)
-> from employee_lavina as E, departments as D
-> where E.department_id = D.department_id group by department_name;
+-----+-----+
| department_name | count(employee_id) |
+-----+-----+
| Engineering     | 2 |
| Sales           | 2 |
| Marketing       | 2 |
+-----+-----+
3 rows in set (0.01 sec)
```